

Nils Matteson

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Inference systems engineer driving cold-start and checkpoint work across vLLM, NVIDIA Dynamo, Alibaba, and Foundry.
8 upstream vLLM PRs merged | **Founder, thaw** (16 PyPI releases) | **Sole author**, inference-systems arXiv paper

Experience

vLLM (vllm-project; open-source fellowship sponsored by Inference)

Jul 2026 – Present

Open-Source Fellow, inference runtime / cold start

Python · C++ · CUDA · PyTorch

- **Roadmap and coordination:** Author and maintainer of #48193; coordinated semantic-readiness work (#46811) and serving-scale startup measurement (#49037). **8 PRs merged**; #51360 approved but unmerged and #54841 in review.
- Built vllm snapshot create/inspect/restore for TP1 initialized engines (#51360): same-host A10 restore reached first correct token in **9.42 s vs 23.91 s** for a warm start (Qwen3-0.6B, n=5, exact-token oracle).
- Designed the pluggable backend behind sleep()/wake_up() (#44074), preserving a no-op path while making CuMem, cuda_checkpoint, and out-of-tree backends composable.
- Diagnosed compiler and kernel-cache correctness failures: #47356 showed --kv-cache-memory-bytes caused a **21–27 s** H100 recompile, and a DeepGEMM content-only key proposal avoided recompiling 41 kernels (**170.6 s miss vs 76.9 s hit**, H100, n=1; #398, open).
- Collapsed a 41-file import refactor into a raw installed-script boundary after maintainer review; #54841 measures CLI feedback from ~3.5 s to ~40 ms (n=5), with no vllm import or server activation.
- **Cross-org technical leadership:** Reviewed NVIDIA Dynamo's vLLM checkpoint integration (#12226), identifying compatibility and failure-path gaps addressed before merge; reviewed Alibaba's #52132 with 15 findings across rollback, health, cleanup, and API compatibility, then used its independent H20 reproduction to harden #51360; validated Foundry SAVE/LOAD, isolated its public-vLLM replay-offset blocker, and landed CUDAGraph export fix #8 (merged).

thaw (open-source LLM inference infrastructure; 16 PyPI releases)

Apr 2026 – Present

Founder & Lead Engineer

Rust · CUDA · Python

- Built snapshot and fork infrastructure for live vLLM/SGLang state; a pre-warmed pool forks in **0.88 s median** on H100 Llama-3.1-8B versus a ~340 s cold boot, amortizing rather than eliminating cold start.
- Streamed cold-cache NVMe restore at **13.0 GB/s** using double-buffered 0_DIRECT/pinned-memory DMA; reconstructed prefix-cache block tables from a KV sidecar and hot-swapped a slot-warm 8B model in **0.29 s at 55 GB/s**.
- TP snapshots are bit-exact on 2×H100 and 2×A40.

Research Cyberinfrastructure, UW–Madison DoIT

Jan 2026 – Apr 2026

AI Workflows Research Assistant

AWS Bedrock · RAG · LLM evaluation

- Built an AWS Bedrock evaluation and cost framework for 9 LLMs, 3 ensembles, and 282 questions; presented *Deploying RAG in Bedrock vs. Local* at UW's ML+X Forum.

Research

Re-feeding Is Not Replaying: Measuring Replay Noise in Counterfactual Token-Credit Estimation

sole author, Jun 2026

arXiv:2606.15621

- Across **3,434 pivots**, 3 models including a GRPO-trained checkpoint, and 6 configurations, re-feeding shifted low-margin token-credit estimates **14–28 percentage points** above replica noise; batch-invariant kernels made all three passes identical. Code, logs, and per-pivot data released.

Technical Writing | Speculative Decoding Engine write-up · Echo and the Engineer

Education

Northeastern University – M.S. Computer Science, admitted; Fall 2026

University of Wisconsin–Madison – B.S. Data Science, Minor in Computer Science

May 2026

Technical Skills

Inference: PyTorch/vLLM internals, torch.compile/Inductor, CUDA graphs, NCCL, CRIU/cuda-checkpoint, Triton, SGLang
Languages and systems: Python, Rust, C++, CUDA; Linux, Docker/BuildKit, Buildkite, gRPC